MIL-E-52891A(ME)
12 July 1978
SUPERSEDING
MIL-E-52891(ME)
21 January 1976

MILITARY SPECIFICATION

ENAMEL, LUSTRELESS, ZINC PHOSPHATE, STYRENATED ALKYD TYPE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a zinc phosphate type of quick-drying synthetic lustreless enamel used as a finishing coat on ammunition and other metal surfaces. It provides a material suitable for use under air pollution regulations.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

FSC 8010

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

PPP-P-1892	- Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
PPP-T-60	- Tape: Packaging, Waterproof.

STANDARDS

FEDERAL

FED. TEST METHOD STD. No. 141	- Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling and Testing.
FED. STD. No. 595	- Colors.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 50 - Chemical Analysis of Yellow, Orange, Red, and Brown Pigments Containing Iron and Manganese.
D 476 - Titanium Dioxide Pigments, Specification For.
D 562 - Consisting of Paints Using the Stormer Viscosimeter, Method of Test For.
D 563 - Phthalic Anhydride Contents of Alkyd Resins and Resin Solutions.
D 610 - Evaluating Degree of Rusting on Painted Steel Surfaces.
D 659 - Evaluating Degree of Chalking of Exterior Paints, Standard Method of.
D 1200 - Viscosity of Paints, Varnishes, and Lacquers by Ford Viscosity Cup, Standard Method of Test For.
D 1210 - Fineness of Dispersion of Pigment-Vehicle Systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

D 1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes.
D 1364 - Water in Volatile Solvents (Fischer Reagent Titration Method).
D 1394 - Chemical Analysis of White Titanium Pigments.
D 1542 - Qualitative Tests for Rosin in Varnishes.
D 2369 - Volatile Content of Paints.
D 2698 - Determination of the Pigment Content of Solvent Type Paints by High Speed Centrifuging, Standard Method For.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

3. REQUIREMENTS

3.1 Qualification. Enamel furnished under this specification shall be products which are qualified for listing on the applicable qualified products list (see 6.4).

3.1.1 Changes. Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under the contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 Color. The enamel shall be furnished in FED. STD. No. 595, color number specified in the contract (see 6.2). When tested as specified in 4.3.8, it shall acceptably match the standard color chip in FED. STD. No. 595.

3.3 Composition.

3.3.1 Vehicle. The vehicle shall be styrenated phthalic alkyd resin together with the necessary amounts of driers and volatile solvents. The resin solution shall have a color value at 50 percent solids content no darker than 10 (Gardner Color Standards of 1953). Small amounts of antioxidants, wetting agents, and stabilizers may be present. The vehicle shall show a

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

negative test for rosin and phenolic resin. The volatile material shall contain no benzene, methanol, chlorinated compounds or other solvent of a highly toxic nature. The volatile solvents used shall conform to the following requirements by volume when tested as specified in 4.3.2:

(a) Aromatic compounds with eight or more carbon atoms except ethyl benzene: 8 percent maximum.
(b) Ethyl benzene and toluene: 20 percent maximum.
(c) Solvents with an olefinic or cyclo-olefinic type of unsaturation: negative test.
(d) Ketones - negative.
(e) Total of (a) plus (b): 20 percent maximum.

Note: Above requirements are in compliance with Southern California Air Pollution Control District, Rule 102.

3.3.2 Pigment. Any combination of the pigments listed in Table I shall make up the basic hiding pigmentation. Hiding pigments shall be chemically pure and free from extenders. The titanium dioxide shall be rutile chalk resisting type conforming to Type III of ASTM D 476. Small amounts of other shading pigments may be used when necessary to match the color chip, provided these additional pigments have good color performance. Zinc phosphate shall constitute 15 $\pm$ 1 percent of the pigment content. Extender pigments shall be siliceous matter, or siliceous matter and barytes, and shall not exceed the amount specified in Table II. The amount of barytes shall not exceed 20 percent of the extender content by weight. Calcium sulfate or carbonate shall not be employed alone or as a component part of any pigment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

Table I. Pigmentation

Color	FED. STD. No. 595 Color No.	Pigmentation
Non-camouflage Forest Green	34079	Yellow Iron Oxide, Medium Chrome Yellow, Phthalocyanine Blue, Carbon or Lampblack.
Olive Drab	34087	Red or Yellow Iron Oxide, Carbon or Lampblack, Medium Chrome Yellow, Titanium Dioxide.
Light Green	34558	Titanium Dioxide, Yellow Iron Oxide, Chrome Yellow, Phthalocyanine Blue or Green, Carbon or Lampblack.
Light Blue	35193	Titanium Dioxide, Yellow Iron Oxide, Phthalocyanine Blue, Carbon or Lampblack.
Blue Gray	36231	Titanium Dioxide, Yellow Iron Oxide, Carbon or Lampblack.
Black	37038	Black Iron Oxide, Carbon or Lampblack.

3.4 Quantitative requirements.

3.4.1 Specific quantitative requirements. Each color shall conform to its specific requirements in Table II when tested as specified in 4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

Table II. Specific Quantitative Requirements

Color Corresponding to Table I	Solids, percent by weight of enamel			Pigment, Percent by Weight of Total Pigment				Dry Opacity
	Total Solids	Pigment Solids		Prime Pigment 1/	Extender pigment	Zinc Phosphate		
	Min.	Min.	Max.	Min.	Max.	Min.	Max.	
Non-camouflage Forest Green	55	34	38	20(Fe_2O_3)	50	14	16	0.98
Olive Drab	55	33	37	2/ 24(Fe_2O_3)	50	14	16	0.98
Light Green	55	36	40	36(TiO_2)	50	14	16	0.98
Light Blue	55	34	38	24(TiO_2)	50	14	16	0.98
Blue Gray	55	34	38	36(TiO_2)	50	14	16	0.98
Black	55	34	38	15(Fe_3O_4)	68	14	16	0.98

1/ On analysis compute prime pigment as indicated in parenthesis.

2/ Lead chromate ($PbCrO_4$) may be substituted on an equal weight basis.

3.4.2 General quantitative requirements. The enamel tested as in 4.3 shall comply with the requirements of Table III.

Table III. General Quantitative Requirements

Characteristics	Requirements	
	Minimum	Maximum
Vehicle solids, percent by weight of enamel	18	--
Phthalic anhydride, percent by weight of vehicle solids	18	26
Rosin and rosin derivatives	Negative	
Phenolic resin	Negative	
Water, percent by weight of enamel	--	1.0
Coarse particles and skins, percent by weight of pigment	--	1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

Table III. General Quantitative Requirements (Cont'd)

Characteristics	Requirements	
	Minimum	Maximum
Viscosity:		
Package, Krebs Stormer shearing rate - 200 rpm:		
Grams	125	175
Equivalent Krebs Units (K.U.)	67	77
Reduced, No. 4 Ford cup, seconds	12	25
Fineness of grind		
Hegman	5	--
ASTM (MICRONS)	--	40
Specular gloss	--	8
Drying time:		
Set to touch, minutes	--	6
Dry hard, minutes	--	10
After tack free, minutes	--	15
Dry through, minutes	--	20
Full hardness, hours	--	72

3.5 Qualitative requirements.

3.5.1 Condition in container. When tested as specified in 4.3.9, the enamel shall be free from grit, seeds, skins, lumps, or livering in a freshly opened full container and shall show no more pigment settling or caking than can be easily and completely reincorporated to a smooth homogeneous state.

3.5.2 Storage stability.

3.5.2.1 Partially full container. When tested as specified in 4.3.10.1, the enamel shall show no skinning. After aging as specified in 4.3.10.1, the enamel shall show no livering, curdling, tough gummy sediment, or hard caking. It shall mix readily to a smooth homogeneous state, and any skin formed shall be continuous and easily removed.

3.5.2.2 Full container. When tested as specified in 4.3.10.2, the enamel shall show no skinning, livering, curdling, hard caking, or tough gummy sediment. It shall remix readily to a smooth homogeneous state, shall have a maximum viscosity of 89 Krebs Units, and shall meet all other requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

3.5.3 Dilution stability. When tested as specified in 4.3.11, the enamel shall remain stable and uniform showing no precipitation, curdling, or separation. Slight pigment settling shall be permitted.

3.5.4 Suspension properties. When tested as specified in 4.3.12, the enamel shall completely redisperse to a smooth homogeneous state.

3.5.5 Brushing properties. When tested as specified in 4.3.13, the enamel shall brush satisfactorily in all respects and shall dry to a smooth, uniform film free from seeds, sags, or streaks.

3.5.6 Dipping properties. When tested as specified in 4.3.14, the enamel shall show satisfactory dipping properties and shall present a smooth appearance free from sagging, running, or excessive silking.

3.5.7 Spraying properties. When tested as specified in 4.3.15, the enamel shall spray satisfactorily in all respects and shall show no running, sagging, streaking, or blushing. The dried film shall show no dusting, mottling, or color separation, and shall present a smooth lusterless finish free from seeds.

3.5.8 Flexibility. When tested as specified in 4.3.16, a film of enamel shall withstand bending without cracking or flaking.

3.5.9 Knife test. When tested as specified in 4.3.17, a film of enamel shall adhere tightly and not flake, crack, or powder from the metal. The cut shall show beveled edges.

3.5.10 Adhesion. When tested as specified in 4.3.18, a film of enamel shall show no removal of the coating by the adhesive tape beyond 1/16 inch on either side of the score line.

3.5.11 Water resistance. When tested as specified in 4.3.19, a film of enamel shall show no wrinkling or blistering immediately after removal of the panel from the water. The enamel shall be no more than slightly affected when examined 2 hours after removal. After 24 hours air drying, the portion of the panel which was immersed shall be almost indistinguishable with regard to hardness, color, and gloss from a panel prepared at the same time but not immersed.

3.5.12 Recoating. When tested as specified in 4.3.20, a film of enamel shall not blister, wrinkle, or show other evidence of lifting. The film shall contain no bleeding pigments which will discolor the white enamel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

3.5.13 Salt spray resistance. When tested as specified in 4.3.21 and examined immediately after removal from the test, a film of enamel shall show no rust creepage or undercutting beyond 1/8 inch from the score mark. At all other points of the panel, there shall be no more than a trace of rusting as defined in ASTM D 610 and no more than five scattered blisters, none larger than 1 mm in diameter. After removal of the enamel, the surface of the steel shall show no more than a trace of rusting, pitting, or corrosion.

3.5.14 Accelerated weathering. When tested as specified in 4.3.22, a film of enamel shall show no more than slight chalking as defined in ASTM D 659 and a color change equivalent to a lightness difference estimate not exceeding 4 units.

3.5.15 Weather resistance. When tested as specified in 4.3.23, a film of enamel shall show no checking, cracking, or film deterioration. There shall be no more than light chalking. The film shall show no excessive change in value or chroma and no change in hue. After removal of any chalking which has occurred, the original color shall be substantially restored and the washed area shall show no more than slight fading or darkening. On removal of the enamel, the surface of the steel shall show no more than a trace of rusting, pitting, or corrosion as defined in ASTM D 610. Rust creepage shall not extend beyond 1/8 inch from the score mark.

3.6 Special markings. Each container shall be legibly marked with the following instructions:

Thinning Directions. For application, the manufacturer shall include on the label thinning recommendations and solvent required.

CAUTION. Flammable liquid. No smoking. During application avoid skin contact and inhalation of vapors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.1.1 Sampling and inspection. Sampling and inspection shall be performed in accordance with Section 1000 of FED. TEST METHOD STD. No. 141.

4.2 Classification of tests. Testing under this specification shall be for the following:

(a) Qualification.
(b) Acceptance of individual lots.
(c) Acceptance for use as component on end item.

4.2.1 Qualification tests. The qualification tests shall consist of tests for all requirements specified in Section 3 (see 6.4).

4.2.2 Acceptance tests. Acceptance tests for acceptance of individual lots shall consist of tests specified in Section 3 with the exception of storage stability (see 3.5.2.2 and 4.3.10.2) and weather resistance (see 3.5.15 and 4.3.23).

4.2.3 Component on an end item. When approved by the cognizant activity, acceptance of lots for use as component on an end item shall be based on compliance with specified requirements for the following characteristics:

Phthalic anhydride
Gloss
Fineness of grind
Flexibility
Knife test
Water resistance
Zinc phosphate

4.3 Test methods.

4.3.1 Test conditions. The routine and referee testing conditions shall be in accordance with Section 7 of FED. TEST METHOD STD. No. 141 except as otherwise specified herein. Failure of any test result to fall within the ranges specified in 3.2, 3.3, 3.4, or 3.5, as applicable, shall constitute failure of the applicable test.

4.3.1.2 Test procedures. The following tests (see Table IV) shall be conducted in accordance with FED. TEST METHOD STD. No. 141, or ASTM, and as specified herein. The right is reserved to make any additional tests deemed necessary to determine that the enamel meets the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

Table IV. Tests and Methods

Test	Applicable method in FED. TEST METHOD STD. No. 141	ASTM	Test Paragraph	Requirement Paragraph
1	2	3	4	5
Isolation of vehicle (super centrifuge)	4032	D 2698	--	3.3.1
Phthalic anhydride...	--	D 563	--	3.4.2
Rosin in isolated vehicle.....	--	D 1542	--	3.4.2
Phenolic resin.....	5141		--	3.4.2
Benzene.....	5091 or 7360	--	--	3.3.1
Methanol.....	5133	--	--	3.3.1
Chlorinated solvents	5132	--	--	3.3.1
Solvent analysis	7355, 7356	--	4.3.2.1	3.3.1
Ketones.....	--	--	4.3.2.2	3.3.1
Total solids.....	4041.1	D 2369	--	3.4.1
Pigment solids.....	4022	D 2698	--	3.4.1
Pigment analysis.....	4021	--	4.3.3	3.4.1
Fe_2O_3, Fe_3O_4 - Iron Oxide.....	7141	D 50	4.3.3.1	3.4.1
$PbCrO_4$ - Lead Chromate	7131	--	4.3.3.2	3.4.1
TiO_2 - Titanium Dioxide.....	7083	D 1394	4.3.3.3	3.4.1
Zinc phosphate...	--	--	4.3.3.4	3.4.1
Other pigments...	--	--	4.3.3.5	3.3.2
Extender pigment, total.....	5271	--	4.3.3.6.1	3.4.1
Extender pigment analysis.....	7281	--	4.3.3.6.2	3.3.2
Dry opacity.....	4122	--	4.3.4	3.4.1
Vehicle solids.....	4041	--	--	3.4.2
Water.....	--	D 1364	--	3.4.2
Coarse particles and skins.....	4092	--	--	3.4.2
Viscosity:				
Package.....	4281	D 562	4.3.5.1	3.4.2
Reduced.....	4282	D 1200	4.3.5.2	3.4.2
Fineness of grind....	4411	D 1210	--	3.4.2
Specular gloss, 60°..	6101	--	4.3.6	3.4.2
Drying time:.....	4061	--	4.3.7	3.4.2
Set to touch.....	4061	--	--	3.4.2
Dry hard.....	4061	--	--	3.4.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

Table IV. Tests and Methods (Cont'd)

1	2	3	4	5
Free from after tack.....	4061	—	—	3.4.2
Dry through.....	4061	—	4.3.7.1	3.4.2
Full hardness.....	--	—	4.3.7.2	3.4.2
Color.....	4250	—	4.3.8	3.2
Condition in container.....	3011	—	4.3.9	3.5.1
Storage stability:...	--	--	4.3.10	3.5.2
Partially full container.....	3021	—	4.3.10.1	3.5.2.1
Full container.....	3022	—	4.3.10.2	3.5.2.2
Dilution stability...	--	--	4.3.11	3.5.3
Suspension properties.....	--	--	4.3.12	3.5.4
Brushing properties..	4321	—	4.3.13	3.5.5
Dipping properties...	4341	--	4.3.14	3.5.6
Spraying properties..	4331, 2131	--	4.3.15	3.5.7
Flexibility.....	6221	—	4.3.16	3.5.8
Knife test.....	6304	—	4.3.17	3.5.9
Adhesion.....	--	—	4.3.18	3.5.10
Water resistance.....	6011	D 1308	4.3.19	3.5.11
Recoating.....	—	--	4.3.20	3.5.12
Salt spray resistance.....	6061, 2011	—	4.3.21	3.5.13
Accelerated weathering.....	6152, 6122	—	4.3.22	3.5.14
Weather resistance...	—	D 610	4.3.23	3.5.15

4.3.2 Solvent analysis.

4.3.2.1 Solvent analysis. Determine the amount of aromatic hydrocarbons, and olefinic and cyclo-olefinic compounds by subjecting the enamel to Test Methods 7355 and 7356 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.3.1 shall constitute failure of this test.

4.3.2.2 Ketones.

4.3.2.2.1 Reagent. The reagent shall be 2 grams of 2,4-dinitrophenyl-hydrazine plus 4 ml of concentrated sulfuric acid plus 30 ml methanol (add slowly) plus 10 ml water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.3.2.2.2 Procedure. Pipette 1 ml of reagent into a 20 x 170 mm test tube. Add 10 drops of distillate and shake for 30 seconds. A yellow precipitate or cloud in the reagent layer indicates the presence of ketones. Run a blank using 1 milliliter of reagent and 10 drops of mineral spirits. Nonconformance to 3.3.1 shall constitute failure of this test.

4.3.3 Analysis of pigment. Extract the pigment as specified in Method 4021 of FED. TEST. METHOD STD. No. 141, using Extraction Mixture C.

4.3.3.1 Iron oxide (Fe_2O_3) content. Determine the iron oxide (Fe_2O_3) content on the extracted pigment in accordance with Method 7141 of FED. TEST METHOD STD. No. 141 or ASTM D 50. Nonconformance to 3.4.1 shall constitute failure of this test.

4.3.3.2 Lead chromate ($PbCrO_4$) content. Determine the lead chromate ($PbCrO_4$) content on the extracted pigment in accordance with Method 7131 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.4.1 shall constitute failure of this test.

4.3.3.3 Titanium dioxide (TiO_2) content. Determine the titanium dioxide (TiO_2) content on the extracted pigment in accordance with Method 7083 of FED. TEST METHOD STD. No. 141 or ASTM Method D 1394. Nonconformance to 3.4.1 shall constitute failure of this test.

4.3.3.4 Zinc phosphate content. Determine the zinc phosphate content in accordance with 4.3.3.4.1 and 4.3.3.4.2.

4.3.3.4.1 Determination of zinc.

4.3.3.4.1.1 Reagents.

(a) Buffer solution (pH 10): 350 ml conc. NH_4OH + 54 g NH_4Cl + H_2O to give 1000 ml.

(b) Eriochrome black T (0.5%): 0.25g eriochrome black T + 2.2g hydroxylamine hydrochloride per 50 ml methanol solution.

(c) Primary standard zinc oxide (0.200N): Accurately weigh 4.069g of oven-dried ZnO. Dissolve it in 250 ml of the buffer solution and dilute to 500.0 ml.

(d) $\frac{N}{5}$ Disodium ethylenediaminetetraacetate dihydrate (EDTA): 37.2g EDTA per liter aqueous solution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.3.3.4.1.2 Procedure.

(a) Accurately weigh approximately 1.0 gram of pigment into a 250 ml glass-stoppered Erlenmeyer flask.
(b) Add 25 ml of buffer, stopper, and shake vigorously every few minutes over a period of 30 minutes.
(c) Filter through fine paper into a 400 ml beaker, washing well with water until 200 ml of filtrate are collected.
(d) Add 20.0 ml of the EDTA (an excess) to the filtrate.
(e) Add 10 drops of eriochrome black T.
(f) Titrate with standard ZnO to a wine-red end point (V_s).
(g) Run a blank by titrating 20.0 ml of the EDTA in 200 ml of an aqueous solution containing 25 ml of the buffer (V_b).

4.3.3.4.1.3 Calculations.

$$\text{percent Zn} = \frac{(V_b - V_s) \times 0.2 \times 3.269}{\text{Sample wt}} \quad (\text{Req} = 6.5 - 7.4\%)$$

$$\text{percent zinc phosphate} = \frac{(V_b - V_s) \times 0.2 \times 7.035}{\text{Sample wt}} \quad (\text{Req} = 14 - 16\%)$$

4.3.3.4.2 Determination of phosphate.

4.3.3.4.2.1 Reagents.

(a) Conc NH_4OH
(b) Conc HNO_3
(c) NG_4NO_3
(d) Ammonium molybdate - Johnson's Formula: Mix 55g of $(NH_4)_6 Mo_7O_{24} \cdot 4H_2O$ and 50g of NH_4NO_3 with 18 ml of conc NH_4OH and 20 ml H_2O. Stir. Dilute to about 700 ml with H_2O, heat with occasional stirring until all salts have dissolved. Dilute to 1000 ml. Let stand overnight. Filter through fine paper but do not wash the residue.

4.3.3.4.2.2 Procedure.

(a) Accurately weigh approximately 2g of pigment into a 250 ml glass-stoppered Erlenmeyer flask.
(b) Add 25 ml of conc. NH_4OH, stopper, and shake vigorously every few minutes over a period of 60 minutes.
(c) Add 25 ml of H_2O and filter through fine paper into a 400 ml beaker, washing well with water.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

(d) Neutralize the filtrate with 7.5N HNO_3 (requires about 35 ml).
(e) Add 15 ml conc. HNO_3 and 6g of NH_4NO_3. Stir.
(f) Heat the clear solution to 80° C (no higher) and add 75 ml of ammonium molybdate with constant stirring.
(g) Stir for several minutes and let the precipitate settle for 2 hours.
(h) Filter through a tared crucible (gooch or medium glass), transfer the precipitate, and wash with 1% HNO_3 5 ml conc. HNO_3 per 500 ml solution). The washing should be thorough.
(i) Give the collected precipitate a final wash with a small amount of water.
(j) Dry the crucible for 2 hours in a 105° C oven.
(k) Cool crucible in a desiccator and determine the weight of the precipitate (it should not exceed 3g; if it does, repeat the determination with a smaller sample).

4.3.3.4.2.3 Calculations.

$$\text{percent } PO_4 = \frac{\text{wt. ppt. x } 5.029}{\text{Sample wt.}} \quad \text{(Req. = 6.3 - 7.2\%)}$$

$$\text{Percent zinc phosphate} = \frac{\text{wt. ppt. x } 11.18}{\text{Sample wt.}} \quad \text{(Req. = 14 - 16\%)}$$

4.3.3.4.3 Failure criteria. Nonconformance to 3.4.1 shall constitute failure of this test.

4.3.3.5 Other pigments. Make appropriate qualitative and quantitative tests on the extracted pigment to determine that only permissible pigments were used in formulating the different colors of enamel. Nonconformance to 3.3.2 shall constitute failure of this test.

4.3.3.6 Extender pigment.

4.3.3.6.1 Total extender pigment content. Determine matter insoluable in acid in the extracted pigment by Method 5271 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.4.1 shall constitute failure of this test.

4.3.3.6.2 Extended pigment analysis. Determine barium sulfate, siliceous material, and calcium (sulfate or carbonate) by the applicable portions of Method 7281 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.3.2 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.3.4 Dry opacity. Draw down the enamel on a black and white Morest card using a film applicator that will deposit a 3-inch-wide film having a dry film thickness of 0.0010 inch maximum. Air dry 72 hours. Determine the reflectance of the film over the black and white portions of the card and calculate the dry opacity. Nonconformance to 3.4.1 shall constitute failure of this test.

4.3.5 Viscosity.

4.3.5.1 Package. Proceed as in Method 4281 of FED. TEST METHOD STD. No. 141 or ASTM Method D 562, except that the method of mixing shall be to agitate the can for 3 minutes on a paint shaker powered by approximately a 1/4-hp motor, operating at about 1350 shakes per minute, manufactured by Red Devil Tools, or equal.

4.3.5.2 Reduced viscosity. Reduce 3 parts by volume of enamel with 1 part by volume of thinner conforming to Table V and determine viscosity in accordance with Method 4282 of FED. TEST METHOD STD. No. 141 or ASTM Method D 1200.

4.3.5.3 Failure criteria. Nonconformance to 3.4.2 shall constitute failure of this test.

Table V. Thinner

Ingredient	Percent by weight
VMP naphtha (8% max aromatic)	65
n-Butyl alcohol	20
Toluene	15

4.3.6 60° specular gloss. Draw down the package material using a 0.002 inch (0.004 inch gap clearance) film applicator. Measure the gloss as specified in Method 6101 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.4.2 shall constitute failure of this test.

4.3.7 Drying time. Determine drying time in accordance with Method 4061 of FED. TEST METHOD STD. No. 141 under referee conditions, except that the enamel shall be drawn down with a 0.002 inch (0.004 inch gap clearance) film applicator. Nonconformance to 3.4.2 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.3.7.1 Dry through. Determine dry through in accordance with Method 4061, except the enamel shall be drawn down with a 0.002 inch (0.004 inch gap clearance) film applicator on a 4 x 12 inch steel panel that has been phosphoric acid etched as in Method 2011, Procedure B of FED. TEST METHOD STD. No. 141.

4.3.7.2 Full hardness. Determine full hardness on a panel prepared as specified in 4.3.7.1. The film shall be considered to have reached full hardness when it is very difficult to remove with a knife blade.

4.3.8 Color. In accordance with Method 4250 of FED. TEST METHOD STD. No. 141, compare the color with the film of enamel on the panel prepared for the hiding power test. Nonconformance to 3.2 shall constitute failure of this test.

4.3.9 Condition in container. Determine package condition for quality conformance testing in accordance with Method 3011 of FED. TEST METHOD STD. No. 141. For qualification testing, determine pigment settling or caking as follows: Process as in Method 3011 of FED. TEST METHOD STD. No. 141, but do not stir. Reseal and then agitate the can for 3 minutes on a paint shaker. On re-examination of the contents, the disclosure of any gel bodies or undispersed pigment indicates unsatisfactory settling properties. Nonconformance to 3.5.1 shall constitute failure of this test.

4.3.10 Storage stability.

4.3.10.1 Partially full container. Determine 48 hour skinning in accordance with Method 3021 of FED. TEST METHOD STD. No. 141 and observe for compliance with 3.5.2.1. Reseal and age for 7 days at 60° C. Nonconformance to 3.5.2.1 shall constitute failure of this test.

4.3.10.2 Full container. In accordance with Method 3022 of FED. TEST METHOD STD. No. 141, allow a full standard quart can of the packaged enamel to stand undisturbed for 6 months and examine the contents. Evaluate pigment settling or caking as in 4.3.9, but agitate the can for 5 minutes on the paint shaker prior to re-examination. Determine viscosity. Nonconformance to 3.5.2.2 shall constitute failure of this test.

4.3.11 Dilution stability. Reduce 1 volume of packaged enamel with 1 volume of thinner conforming to Table V by slowly adding the thinner to the enamel with constant stirring. Allow to stand 24 hours, thoroughly remix, and, using a 0.0005 inch (0.0010 inch gap clearance) film applicator, draw down a film of the mixture on clear plate glass. Before and after the film is dry, examine by transmitted light. Nonconformance to 3.5.3 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.3.12 Suspension properties. Reduce the enamel as specified in 4.3.5.2. Place 6 ounces of the reduced enamel in an 8-ounce glass jar. Allow the stoppered jar to remain undisturbed for 24 hours and then place the unopened jar on a paint shaker (see 4.3.5.1) and agitate the contents for 20 seconds. Re-examine the material for any evidence of nonhomogeneity or undispersed pigment. Nonconformance to 3.5.4 shall constitute failure of this test.

4.3.13 Brushing properties. Apply the enamel to a 4 x 12 inch steel panel, using a 1-1/2 inch brush, and observe for brushing properties in accordance with Method 4321 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.5.5 shall constitute failure of this test.

4.3.14 Dipping properties. Reduce 3 parts by volume of enamel with 1 part by volume of thinner conforming to Table V. Test the dipping properties of the thinned enamel in accordance with Method 4341 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.5.6 shall constitute failure of this test.

4.3.15 Spraying properties. Reduce the enamel as specified in 4.3.5.2. Spray on a steel panel to give a dry film thickness of 0.0009 to 0.0011 inch and observe for spraying properties in accordance with Method 4331 of FED. TEST METHOD STD. No. 141. For referee test, use automatic application per Method 2131 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.5.7 shall constitute failure of this test.

4.3.16 Flexibility. Determine flexibility in accordance with Method 6221 of FED. TEST METHOD STD. No. 141. Draw down a 2-inch-wide film of enamel with suitable film applicator that will give a dry film thickness of 0.0009 to 0.0011 inch on a smooth finish steel panel prepared in accordance with Method 2012 of FED. TEST METHOD STD. No. 141, using the petroleum naphtha ethylene glycol monoethylether mixture. Allow the test panel to air dry 1/2 hour and then bake for 24 hours at 105° ± 2° C (221° ± 4° F). Condition the panel for 1/2 hour at 25° ± 1° C. Bend over a 1/4 inch mandrel. Nonconformance to 3.5.8 shall constitute failure of this test.

4.3.17 Knife test. Perform the knife test in accordance with Method 6304 of FED. TEST METHOD STD. No. 141 using a flat portion of the baked panel from the flexibility test specified in 4.3.16. Nonconformance to 3.5.9 shall constitute failure of this test.

4.3.18 Adhesion, tape test.

4.3.18.1 Panel preparation. Using a 0.0020 inch (0.0040 inch gap clearance) film applicator, draw down a 2-inch-wide film of the enamel on a steel panel, cleaned with solvent and etched with phosphoric acid as specified in Procedure B, Method 2011 of FED. TEST METHOD STD. No. 141.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

4.3.18.2 Procedure. Air dry the specimen for 1 hour under referee conditions and then score a line through to the metal across the width of the film using a sharp pointed knife. The film shall then be taped perpendicular to and across the score line with waterproof, pressure-sensitive adhesive tape 3/4 inch wide, conforming to PPP-T-60, Type IV. The tape shall be pressed in firm contact with the film and shall extend for approximately 1 inch on each side of the score line. All air bubbles shall be rolled out by firm pressure of the thumb. Allow approximately 10 seconds for the test area to return to room temperature. Grasp a free end of the tape and, at a rapid speed, strip it from the specimen by pulling the tape back upon itself at an angle of 180 degrees. Nonconformance to 3.5.10 shall constitute failure of this test.

4.3.19 Water resistance. Prepare two panels as specified in 4.3.18.1 and air dry for 72 hours. Coat all exposed, uncoated metal surfaces with wax or other suitable coating and immerse one of the panels for 18 hours in distilled water at 23° ± 1° C (73.4° ± 2° F) in accordance with Method 6011 of FED. TEST METHOD STD. No. 141 or ASTM D 1308. At the end of the test period, remove the panel from the water. Nonconformance to 3.5.11 shall constitute failure of this test.

4.3.20 Recoating. Prepare two panels as specified in 4.3.18.1 and air dry for 24 and 72 hours, respectively. At the end of its drying period, immerse each panel to a depth of 2-1/2 inches in enamel which has been reduced with 1 part by volume of thinner conforming to Table V to 4 parts by volume of package material. At the end of 5 seconds, remove the panel, dry in a vertical position. Nonconformance to 3.5.12 shall constitute failure of this test.

4.3.21 Salt spray resistance. Solvent clean three 4 x 12 inch steel panels in accordance with Method 2011 of FED. TEST METHOD STD. No. 141 and apply a phosphate coating conforming to TT-C-490, Type I. Reduce the enamel as specified in 4.3.5.2 and spray the test panels to a uniform dry film thickness of 0.0009 to 0.0011 inch. Air dry for 72 hours, score, and expose to 5 percent salt spray for 150 hours as specified in Method 6061 of FED. TEST METHOD STD. No. 141. Upon removal, wash the panels gently in running water not warmer than 100° F until free from any visible salt deposits. Strip the enamel film from the panels by means of lacquer thinner and inspect the steel for rust, pitting, or corrosion. Nonconformance to 3.5.13 shall constitute failure of this test.

4.3.22 Accelerated weathering. Prepare a film of the enamel as specified in 4.3.16 and air dry for 72 hours. Measure the directional reflectance and expose the panel to accelerated weathering for 168 hours in accordance

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

with Method 6152 of FED. TEST METHOD STD. No. 141, using a twin arc apparatus. Examine the exposed panel for chalking by rubbing with a piece of velvet or cheesecloth wrapped around the finger. Using moderate pressure, draw the cloth across the width of the panel in two different directions. Measure the directional reflectance (Method 6121) on an unrubbed area of the exposed panel and determine the amount of color change, expressed as lightness index difference (ΔL), using Method 6122 of FED. TEST METHOD STD. No. 141. Nonconformance to 3.5.14 shall constitute failure of this test.

4.3.23 Weather resistance. Prepare two scored test panels as specified in 4.3.21. Air dry for 72 hours and place on outdoor exposure for 18 months at an angle of 45 degrees south in the climate of Washington, DC. At the conclusion of the exposure period, inspect the panels. Determine chalking as specified in 4.3.22. Wash the panels with a warm soap solution using a soft sponge or cloth, rinse dry, and examine for color change. Strip the enamel from the panels by means of lacquer thinner and inspect the steel for rust, pitting, or corrosion. Nonconformance to 3.5.15 shall constitute failure of this test.

4.4 Inspection of packaging. The preservation, packing, and marking shall be examined for compliance with the quality assurance requirements of PPP-P-1892 for Level A and B.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial as specified (see 6.2).

5.1.1 Level A and B. The enamel shall be preserved in accordance with PPP-P-1892 as specified therein for packaging. The enamel shall be furnished in 1-quart or 1-gallon multiple friction top containers, in 5-gallon lug cover steel pails, or in 55-gallon steel drums as specified (see 6.2).

5.1.2 Commercial. The enamel shall be preserved in accordance with MIL-STD-1188. The enamel shall be furnished in 1-quart or 1-gallon multiple friction top containers, in 5-gallon lug cover steel pails, or in 55-gallon steel drums, as specified (see 6.2).

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A and B. The enamel shall be packed in accordance with PPP-P-1892.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

5.2.2 Commercial. The enamel shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Level A and B. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The enamel covered by this specification is intended for use as a finish coat on phosphated or primed metal surfaces and on ammunition components. IT IS NOT INTENDED for use where gasoline resistance is a prime requisite such as automotive equipment.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color and color number (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) Size of container required (see 5.1 and 5.2).

6.3 The enamel covered by this specification should be purchased by volume, the unit being 1 U.S. liquid gallon of 231 cubic inches at 68° F (20° C).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable qualified products list whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Commanding Officer, US Army Mobility Equipment Research and Development Command, Fort Belvoir, VA 22060, ATTN: DRDME-VO and information pertaining to qualification of products may be obtained from that activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

6.5 Enamels of this type show some tendency to be thixotropic at higher viscosities. The use of a material such as diethylamine to the extent of 0.5 to 1.0 percent on the resin solids basis has been found to reduce this condition materially and improve the package stability.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
Army - MR

Project 8010-A135

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment Research & Development Command
ATTN: DRDME-DS
Fort Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil – Downloaded: 2026-10-22 12:13:02
Check the source to verify that this is the current version before use.